AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T103/TM103
Computer Organization and Architecture

Midterm Examination (MTA) –Makeup
Spring Semester 2025/2026

Date: 30/Apr./2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 90 mins

Instructions:

- Total Marks: **30**
- This exam consists of **TWO** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Programmable Calculators** are **NOT** allowed.

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Part 1: Multiple Choices questions [5 Marks]

Q-1: Answer all the following questions by choosing the most correct statement.

1- The largest decimal number that can be represented in a **6-bit unsigned** whole numbering system is:
   a. 31.
   b. 32.
   c. 63.
   d. 64.

2- What is the best answer that describes the XOR operator?
   a. The output is 0 when we have even number of 1's at the input.
   b. The output is 1, if any of the inputs is 1.
   c. The output is 1, only if all inputs are 1.
   d. Both a and c

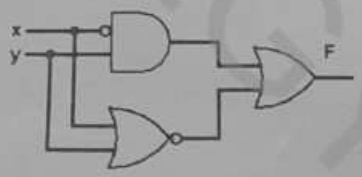
3- Computer speed is mainly affected by:
   a. RAM
   b. I/O Devices
   c. CPU's Clock Speed
   d. VGA Card.

4- How many milliseconds (ms) are in 7 microseconds?
   a. $7 * 2^{10}$.
   b. $7 * 10^{-3}$.
   c. $7 * 10^{3}$.
   d. $7 * 2^{-10}$.

5- Which of the following is not true about Decoders?
   a. They have n inputs and 2^n outputs.
   b. They are sequential.
   c. Only one output is active at a time, based on the value present at the input lines.
   d. None of the above

PART 2: ALL QUESTIONS ARE REQUIRED [25 Marks]

Q-2: Given the circuit below that implements the Boolean expression F. **[2 marks]** What will the value at the output be, if the input values are: x=1, y=0? Justify your answer.

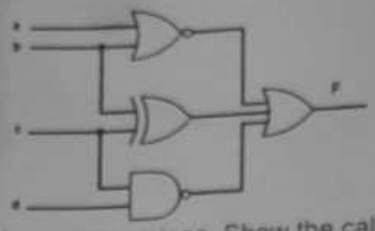


Q-3: Draw the truth table of the NAND and NOR gates, for x and y inputs. **[2 marks]**

Q-4: Below is the combinational circuit that directly implements the Boolean expression F. Write the equation of F. **[3 marks]**

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Q-5: Perform the following base conversions. Show the calculation steps: **[3 marks]**

a. $(175.75)_{10} = (\quad)_{16}$ [1.5 marks]

b. $(163.2)_8 = (\quad)_{10}$ [1.5 marks]

Q-6: Answer the following questions: **[3 marks]**

a- What is a cache memory?

b- Name two types of ports in a computer (whether old or new)

c- Name two levels of the computer hierarchy

Q-7: Represent the decimal number (**-79**) in: **[5 marks]**

a) Signed Magnitude 8-bit representation. (1 mark)

b) Two's Complement 8-bit representation. (2 marks)

c) Excess-16 Floating point representation. (2 marks)

Q-8: Given the following truth table: **[7 marks]**

a) Deduce from the above table, the Sum of Product form of F.

b) Draw the K-map of function **F**, showing the groupings.

c) Deduce the simplified F from the K-map.

Inputs			Output
a	*b*	*c*	*F*
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

End of questions

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